

Part: 3 Feature Selection | Embedded methods

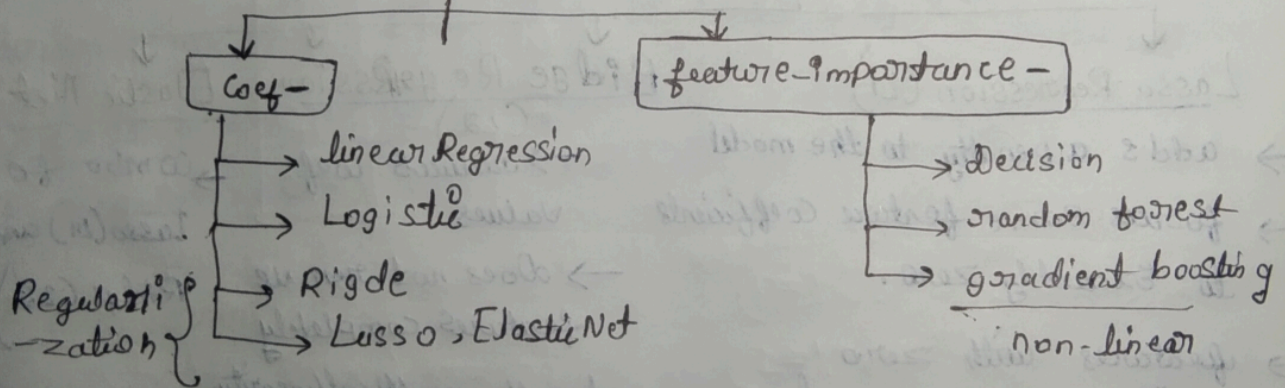
Embedded methods are feature selection techniques where feature selection happens while the model is being trained

- No separate feature selection step
- The model itself decides which features are important
- The model learns and selects important features during training.

How Embedded methods work

- The model is trained on all features
- During training:
 - Important features get higher weight
 - Unimportant features get very small or zero weight
- Features with zero or low importance can be removed.

Embedded methods
↓
machine learning model



i) Linear Regression :-

cgpa | iq | lpa

$$lpa = \beta_0 + \boxed{\beta_1} \text{cgpa} + \boxed{\beta_2} \text{iq}$$

weight or importance

Coef- ⇒ Feature Importance

Linear Regression has some following assumptions, ~~There~~ If these assumptions are failed, Then there will be no meaning to extract these coefficients.

i) Linearity :- The relationship between the independent and dependent variables is linear.

ii) Independence :- The observations are independent of each other, this implies that the residuals (the difference b/w the observed & predicted values) are independent.

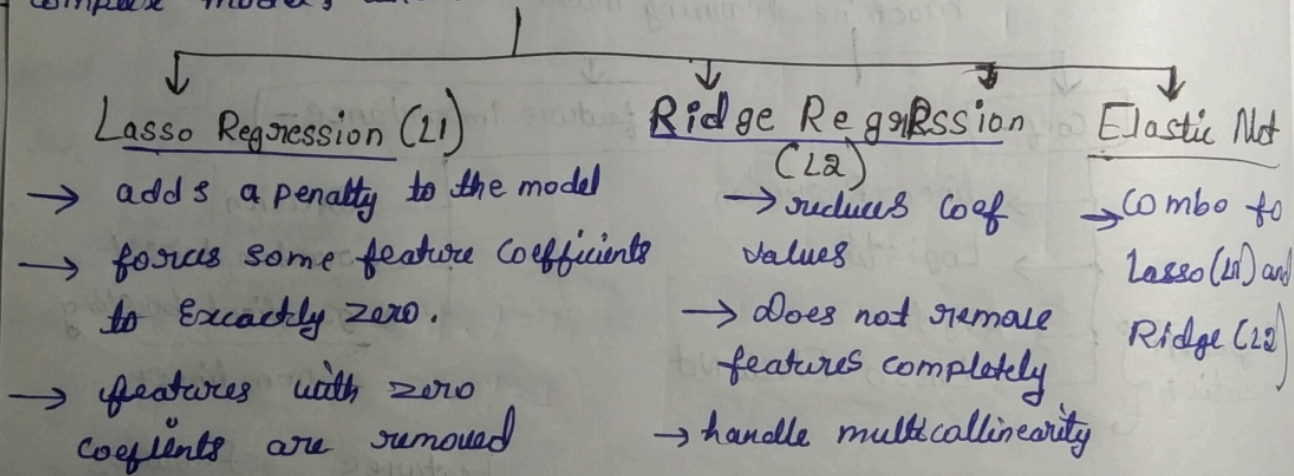
iii) Homoscedasticity :- The variance of the residuals is constant across all levels of the independent variables.

iv) Normality :- The residuals are normally distributed.

v) No multicollinearity :- Independent variables are not highly correlated with each other. This assumption is very important when you want to interpret regression coefficients.

Regularized Models :-

Regularized linear models are linear models that include a penalty term in the loss function during training.
→ This penalty term discourages the learning of a too complex model, which can help in preventing overfitting.



Tree-Based Models :-

Decision Trees, Random Forest, Gradient Boosting.

→ works well for non-linear data.

Advantages

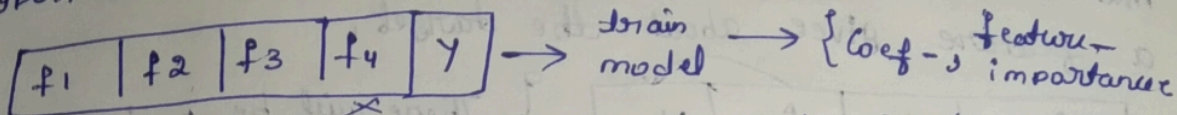
- Faster than wrapper methods
- Better than filter methods
- Less overfitting
- Used widely in real projects

Disadvantages

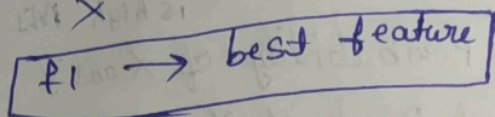
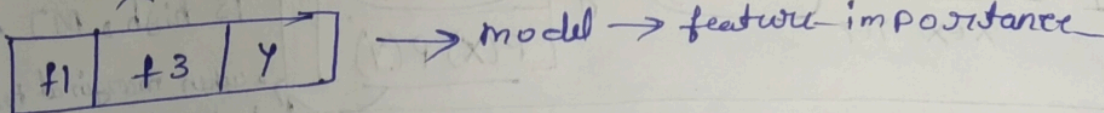
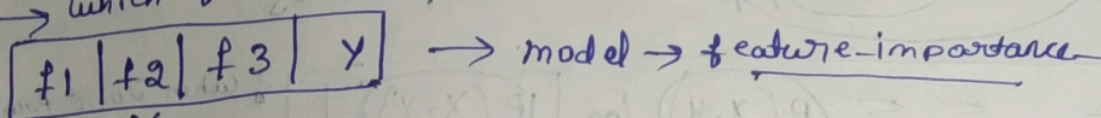
- Model - dependent
- Feature Importance may vary by model.

Recursive Feature Elimination (RFE)

Hybrid $\rightarrow$ { wrapper, Embedded methods }



$\rightarrow$ which feature has low importance, that will be dropped



advantages OF Embedded methods:-

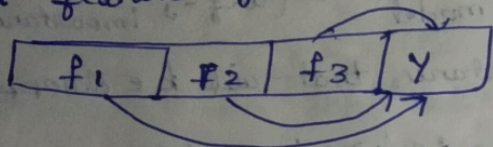
- Performance: They are generally more accurate than filter methods, since they interact between features.
- Efficiency: They are more computationally efficient, than wrapper methods since they fit model only once.
- Less Prone to Overfitting: They introduce some form of regularization, which helps to avoid overfitting.

Disadvantages:-

- Model Specific: They are tied to a specific machine learning model, the selected features are not necessarily optimal for other models.
- Complexity: They can be more complex and harder to interpret than filter methods.
 $\rightarrow$ For example: understanding why Lasso shrinks some coefficients to zero and not others.
- Tuning required: hyperparameter like (threshold value) need to tune carefully.
- Stability: depending on the model & the data, small change in data can result in different set of selected features.

II. Mutual Information | Filter methods

Mutual Information measures how much information a feature gives about the target.



$$MI = \sum_{X \in X} \sum_{Y \in Y} P(X, Y) \log \left[\frac{P(X, Y)}{P(X)P(Y)} \right]$$

• when Joint prob is high, MI is high
• when marginal prob is high, MI is low

where, $P(X, Y)$ → Joint Probability of X and Y

$P(X)$ → Marginal Prob of X

$P(Y)$ → Marginal Prob of Y

Example:-

(X)	Sex	Survived (Y)
	M	0 ✓
	F	1
	M	0 ✓
	F	0 ✓
	M	1

		Survived		
		0	1	
Sex	M	2/5	1/5	3/5
	F	1/5	1/5	2/5
		3/5	2/5	

Joint Prob
Marginal values

$$2/5 \Rightarrow (X=M, Y=0)$$

$$1/5 \Rightarrow (X=F, Y=0)$$

$$1/5 \Rightarrow (X=M, Y=1)$$

$$1/5 \Rightarrow (X=F, Y=1)$$

From Formula:-

$$MI = \frac{2}{5} \log \left(\frac{2/5}{3/5 \times 3/5} \right) + \frac{1}{5} \log \left(\frac{1/5}{3/5 \times 2/5} \right) + \frac{1}{5} \log \left(\frac{1/5}{2/5 \times 3/5} \right) + \frac{1}{5} \log \left(\frac{1/5}{2/5 \times 2/5} \right)$$

→ which tells importance of importance of Sex (X) in order to calculate Survived (Y)

→ It is Symmetric :- $MI(X, Y) = MI(Y, X)$

→ It can capture any kind of relationship including nonlinear one

Advantages of MI:

- i) Captures non-linear relationships: Unlike correlation, MI can detect any type of dependency (linear & non linear).
- ii) Works for Categorical and regression: applicable to both continuous and categorical variables.
- iii) Model Independent (fitter method): does not depend on any ML model.
- iv) No assumptions of linearity and Normality distribution.
- v) Good for complex data → when relationships are unknown.

Disadvantages of MI:

- i) Computationally Expensive: estimating MI is slow for large datasets (too many cols).
- ii) Difficult to interpret: MI values do not indicate direction (+ve, -ve).
- iii) Sensitive to estimation method: results depending on binning or k-nearest neighbor estimation.

binning ⇒ histogram

0-20	21-50	51-80
# 1	# 2	# 3
35 (2)	60 (3)	52 (3)
16 (1)	27 (2)	

if age	Survived
35 (2)	0
60 (3)	1
52 (3)	0
16 (1)	1
27 (2)	1

- iv) Cannot handle feature redundancy well: may select highly correlated features together.
- v) Does not Consider Feature Interaction with others: evaluates Each feature individually.